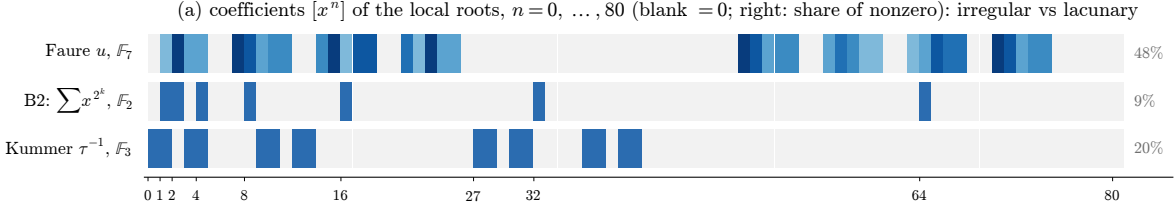
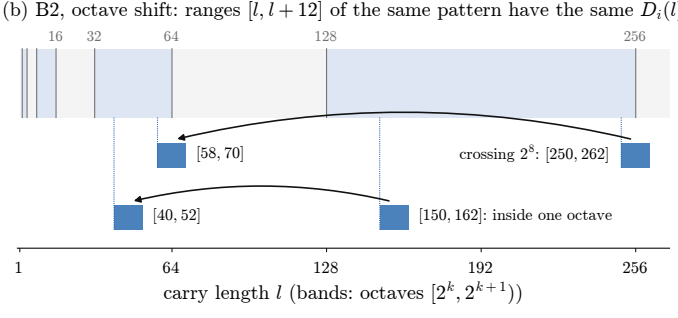
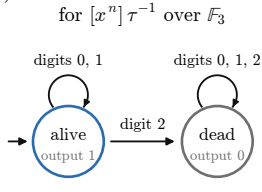

(a) coefficients $[x^n]$ of the local roots, $n = 0, \ldots, 80$ (blank $= 0$; right: share of nonzero): irregular vs lacunary
Faure u, $\mathbb{F}_7$
B2: $\sum x^{2^k}$, $\mathbb{F}_2$
Kummer τ^{-1}, $\mathbb{F}_3$
48%
9%
20%
0 1 2 4 8 16 27 32 64 80


(b) B2, octave shift: ranges $[l, l+12]$ of the same pattern have the same $D_i(l)$
16 32 64 128 256
[58, 70]
crossing 2^8: [250, 262]
[40, 52]
[150, 162]: inside one octave
1 64 128 192 256
carry length l (bands: octaves $[2^k, 2^{k+1})$)


(c) schematic: two-state automaton for $[x^n]\,\tau^{-1}$ over $\mathbb{F}_3$
digits 0, 1
digits 0, 1, 2
alive
output 1
digit 2
dead
output 0
e.g. $4 = 11_3 \mapsto 1$, $5 = 12_3 \mapsto 0$